2016 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.306 Internet Applications Engineering

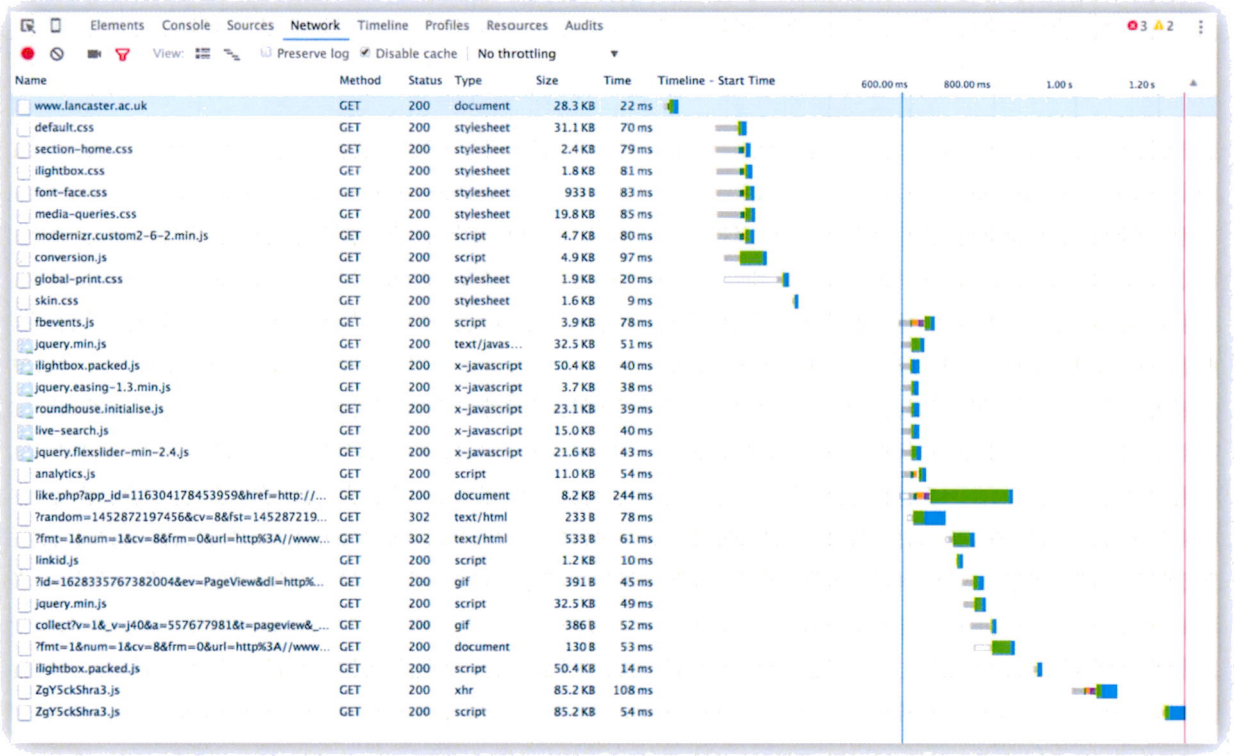
Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

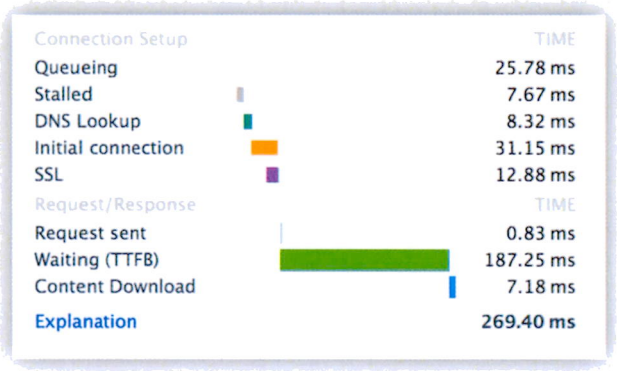
Use a separate answer book for each question.

Question 1

This question is about analysing web site performance and improving web site scalability.

1.a The following images are taken from Google Chrome Developer Tools. They show a network trace taken from the 'Lancaster University' web site and then a more detailed timing breakdown of the 'like.php' request.





i. The first image shows blue (leftmost) and red (rightmost) vertical lines at approximately 0.57 and 1.26 seconds respectively. Explain what the two lines represent, and their significance for website performance.
Some of the JavaScript files are loaded *before* the blue line and some loaded *after*. What might explain this difference in behaviour between the two?

[6 marks]

Question 1 Continues on Next Page...

Question 1 Continued…

ii. Explain in detail what is conveyed by each stage of the detailed request breakdown.

a) Queueing
b) Stalled
c) DNS Lookup
d) Initial connection
e) SSL
f) Request Sent
g) Waiting (TTFB)
h) Content Download

[7 marks]

iii. Looking at the timing breakdown of the like.php resource, what can you infer from the time spent in each of the categories? What might be the explanation for any delay observed?

[4 marks]

iv. In general, how will the retrieval process differ on a subsequent refresh of the webpage? How might this effect the performance?

[2 marks]

1.b A website owner starts to identify performance challenges when their site is visited by an increasingly large user base. What two approaches to scaling could they use to accommodate this growing user base? Discuss the trade-offs between these two techniques.

[6 marks]

Total 25 marks

Question 2

Web analytics provide mechanisms for tracking viewer interactions.

2.a Describe the following terms and explain in detail why a webpage might have high values for each one of these as well as what this might mean for the owner of the website:

(i) Bounce rate.
(ii) Exit rate.
(iii) Conversion rate.
(iv) Engagement rate.

[8 marks]

2.b Consider the emergence of wearable technologies such as the Apple Watch or Samsung Gear. Describe in detail the type of analytics data one might want to collect about the use of such systems.

[4 marks]

2.c Explain the distinction between the terms 'hits', 'visits' and 'visitors' in relation to a website. Using scenarios explain why is it particularly challenging to determine the number of 'visitors' to a website?

[6 marks]

2.d What privacy concerns does analytics raise? Consider the type of information captured and the potential implications this has on privacy.

[7 marks]

Total 25 marks

Question 3

This question is about responsive web design and continuous delivery.

3.a The following three principles are considered core to responsive web design. Describe how each is implemented and how they benefit a responsive website.

1. Fluid Images
2. Fluid Grids
3. Media Queries

[6 marks]

3.b Look at the following CSS code snippet that uses a responsive design technique to resize an image. With higher resolution devices the image should take up a proportionately smaller part of the screen.

```
img {
      max-width: 100%
}

@media(min-width:700px, max-width:1000px){
      img {
            max-width: 50%;
      }
}

@media(min-width:1000px, max-width:1500px){
      img {
            max-width: 33%
      }
}
```

i. What would be the problem with the current code snippet for certain devices?

[1 mark]

ii. Modify the code to show how this problem could be resolved.

[2 marks]

iii. The above code snippet takes an approach whereby different designs are served to devices based on their screen pixel width. Discuss three device characteristics that could be used trigger the selection of an alternative design and describe a use case scenario for each of these.

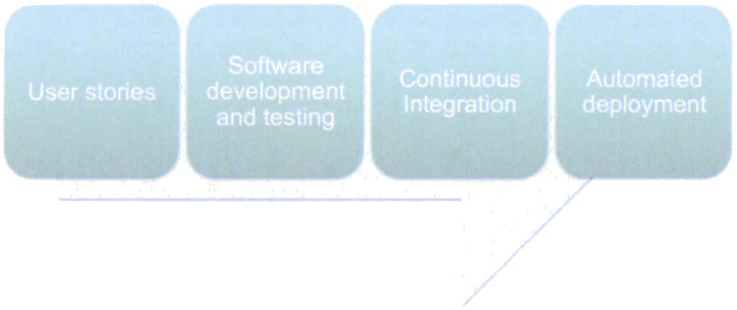
[6 marks]

Question 3 continues on next page...

Question 3 continued…

3.c Continuous delivery is a software engineering approach. What benefits does it provide to an organisation?

[2 marks]



3.d The image above shows the continuous delivery pipeline. Explain the purpose of each stage in the pipeline and any requirements that would be necessary in order to move onto the next stage.

[4 marks]

3.e Consider the BBC news website, which has journalists that publish news stories and visitors that read the news.

Write two user stories, one from the point of a journalist that wishes to publish an article, and another from the perspective of a visitor to the website.

[4 marks]

Total 25 marks

Question 4

The following question is about web performance best practices and HTTP/2.

4.a Caching is traditionally used by website owners (such as the BBC) in order to scale their infrastructure to being capable of handling millions of website views per day.

i. Describe how and where caching is used within a traditional website architecture, using a diagram to assist your answer.

[4 marks]

ii. Explain how the use of caching introduces challenges when architecting systems capable of supporting interactive web experiences.

[2 marks]

iii. You have been asked to design a system architecture and propose a set of technologies for a website capable of supporting interactive communications. Based on your understanding of websites with time sensitive interactions, outline how you might approach this. Give an overview of the system architecture and the technologies you believe would play a key role, including a rationale for their selection. What impact does your architecture and technology choices have on the computational resources required to provide the service?

[6 marks]

4.b Discuss in detail what is meant by the following statement:

"Web performance best practices are effectively working around the core deficiencies of HTTP/1.x"

[5 marks]

4.c HTTP/2 has been designed to improve web performance by making more efficient use of network resources and reducing the perception of latency. Describe four techniques that HTTP/2 uses and explain how each of them achieves an improvement in performance.

[8 marks]

Total 25 marks

---End of Paper---